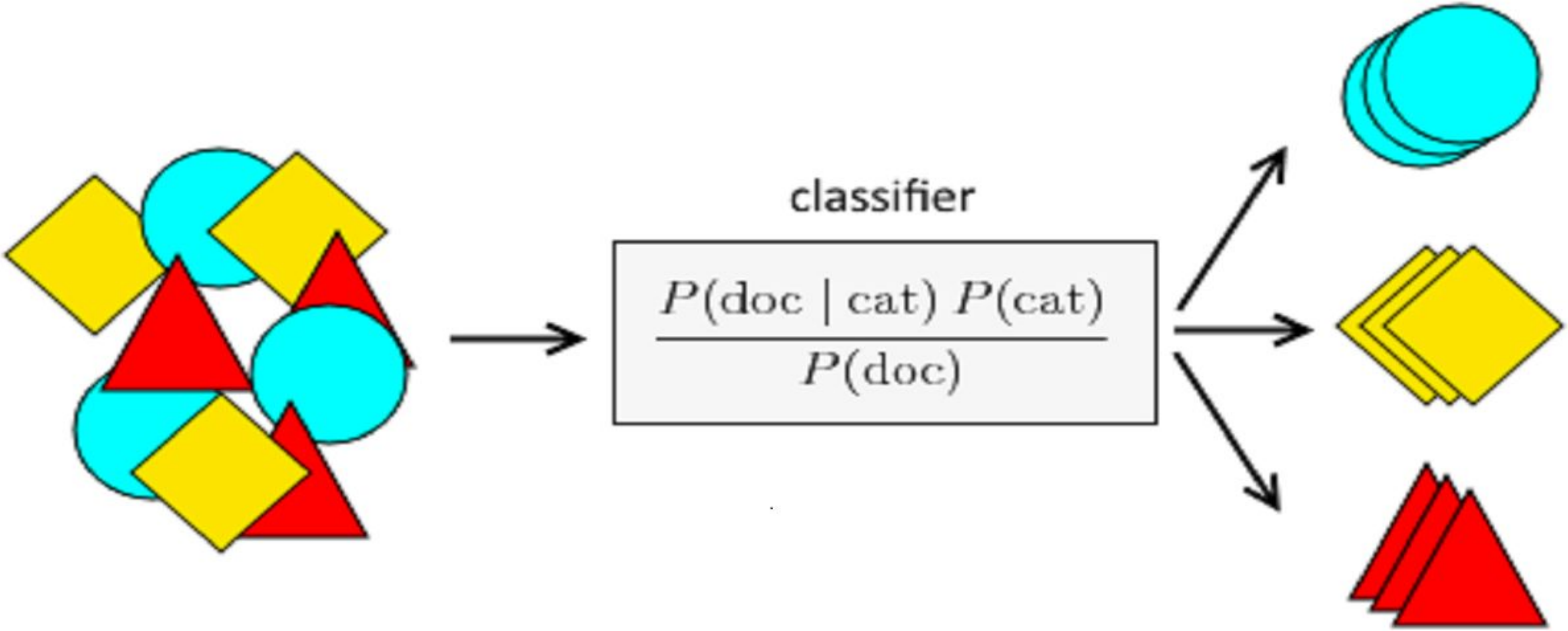
Naive Bayes Classification



Naive -Bayes

Naïve: It is called Naïve because it assumes that the occurrence of a certain feature is independent of the occurrence of other features.

Such as if the fruit is identified on the bases of color, shape, and taste, then red, spherical, and sweet fruit is recognized as an apple.

Bayes: It is called Bayes because it depends on the principle of Bayes' Theorem.

Naive Bayes Algorithm

The technique descended from the work of the 18^{th} century mathematician Thomas Bayes, who developed foundational mathematical principles (now known as Bayesian methods) for describing the probability of events, and how probabilities should be revised in light of additional information.

Bayesian Classifier-Applications

- Bayesian classifiers have been used for:
- **Text classification,** such as junk email (spam) filtering, author identification,or topic categorization
- **Intrusion detection** or anomaly detection in computer networks
- **Diagnosing medical conditions,** when given a set of observed symptoms

Bayesian probability theory-Concepts

- Bayesian probability theory is rooted in the idea that the estimated likelihood of an event should be based on the evidence at hand.
- **Events** are possible outcomes, such as sunny and rainy weather, a heads or tails result in a coin flip, or spam and not spam email messages.
- A **trial** is a single opportunity for the event to occur, such as a day's weather, a coin flip, or an email message

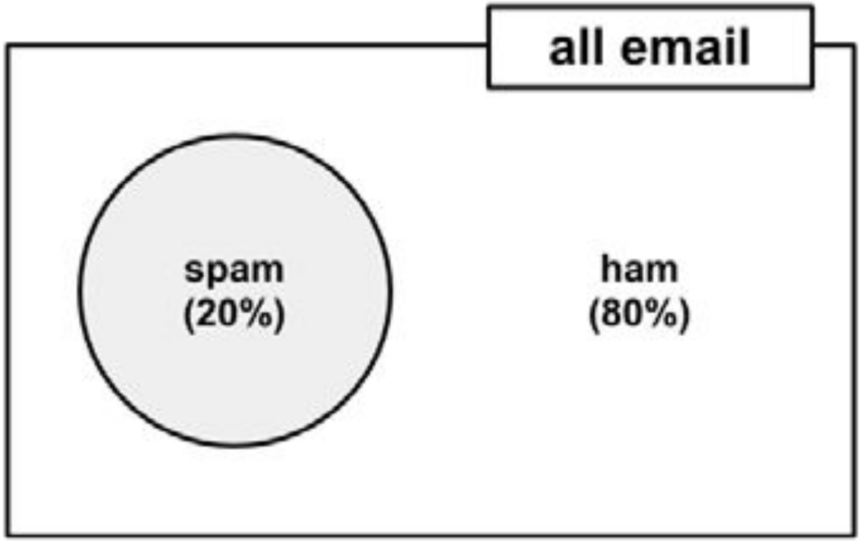
Probability

- The probability of an event can be estimated from observed data by dividing the number of trials in which an event occurred by the total number of trials.
- For instance, if it rained **3 out of 10** days, the probability of rain can be estimated as **30 percent.**
- Similarly, if **10 out of 50 email messages are spam**, then the probability of spam can be estimated as **20 percent**. The notation P(A) is used to denote the
- probability of event A, as in **P(spam) = 0.20**
- **The total probability of all possible outcomes of a trial must always be 100 percent.**

- If the trial only has two outcomes that cannot occur simultaneously, such as heads or tails, or spam and ham (non-spam), then knowing the probability of either outcome reveals the probability of the other.
- For example, given the value
- P(spam) = **0.20**, we are able to calculate
- P(ham) = **1 – 0.20 = 0.80.**

Mutually Exclusive events

- This works because the events spam and ham are **mutually exclusive and exhaustive**. This means that the events cannot occur at the same time and are the only two possible outcomes.
- As shorthand, the notation P(¬A) can be used to denote the probability of event A not occurring, as in **P(¬spam) = 0.80.**


all email
spam
(20%)
ham
(80%)

- Joint Probability: Probability of two (or more) simultaneous events, e.g. P(A and B) or P(A, B).
- The conditional probability **is the probability of one event given the occurrence of another event**, often described in terms of events A and B from two dependent random variables e.g. X and Y.

- Conditional Probability: Probability of one (or more) event given the occurrence of another event, e.g. P(A given B) or P(A | B).

Conditional probability with Bayes' theorem

- The relationships between dependent events can be described using Bayes' theorem, as shown in the following formula. The notation $P(A|B)$ can be read as the probability of event A given that event B occurred.
- This is known as conditional probability, since the probability of A is dependent (that is, conditional) on what happened with event B.

Bayesian Formula

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}$$

Decision Rule Using Conditional Probabilities

Likelihood

Class Prior Probability

$$P(c \mid x) = \frac{P(x \mid c)\,P(c)}{P(x)}$$

Posterior Probability

Predictor Prior Probability

$$P(c \mid X) = P(x_1 \mid c) \times P(x_2 \mid c) \times \cdots \times P(x_n \mid c) \times P(c)$$

Patient	Fever	Cough	Body Ache	Fatigue	Disease
1	Yes	Yes	No	Yes	Disease
2	No	Yes	Yes	No	No Disease
3	Yes	No	Yes	Yes	Disease
4	Yes	Yes	Yes	No	No Disease
5	No	Yes	No	No	No Disease
6	Yes	Yes	Yes	Yes	Disease
7	Yes	No	No	Yes	Disease
8	No	No	Yes	No	No Disease
9	Yes	Yes	No	Yes	Disease
10	No	No	No	No	No Disease

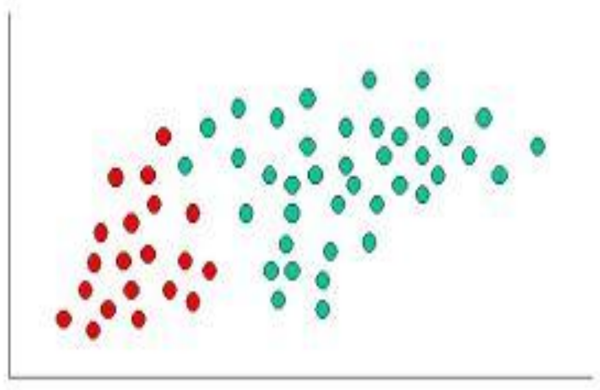
- $P(c|x)$ is the posterior probability of *class* (c, *target*) given *predictor* (x, *attributes*).
- $P(c)$ is the prior probability of *class*.
- $P(x|c)$ is the likelihood which is the probability of the *predictor* given *class*.
- $P(x)$ is the prior probability of the *predictor*.

$$P(y|X) = \frac{P(X|y) * P(y)}{P(X)}$$

CLASSIFY NEW OBJECT :DECIDE TO WHICH CLASS LABEL THEY BELONG, RED OR GREEN

Green objects =40

Red objects =20

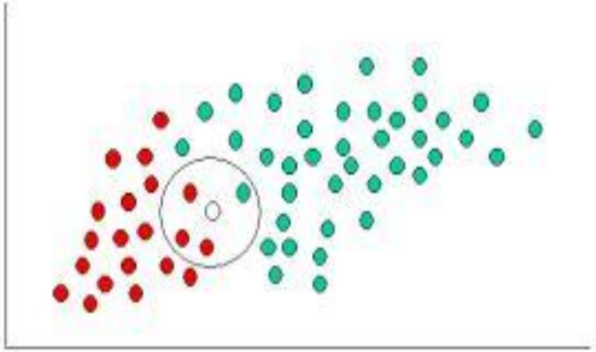


$$\text{Prior probability for GREEN} \propto \frac{\text{Number of GREEN objects}}{\text{Total number of objects}}$$

$$\text{Prior probability for RED} \propto \frac{\text{Number of RED objects}}{\text{Total number of objects}}$$

$$\text{Prior probability for GREEN} \propto \frac{40}{60}$$

$$\text{Prior probability for RED} \propto \frac{20}{60}$$



$$\text{Likelihood of } X \text{ given GREEN} \propto \frac{\text{Number of GREEN in the vicinity of } X}{\text{Total number of GREEN cases}}$$

$$\text{Likelihood of } X \text{ given RED} \propto \frac{\text{Number of RED in the vicinity of } X}{\text{Total number of RED cases}}$$

$$\text{Probability of } X \text{ given GREEN} \propto \frac{1}{40}$$

$$\text{Probability of } X \text{ given RED} \propto \frac{3}{20}$$

$$\text{Posterior probability of } X \text{ being GREEN} \propto$$
$$\text{Prior probability of GREEN} \times \text{Likelihood of } X \text{ given GREEN}$$
$$= \frac{4}{6} \times \frac{1}{40} = \frac{1}{60}$$

$$\text{Posterior probability of } X \text{ being RED} \propto$$
$$\text{Prior probability of RED} \times \text{Likelihood of } X \text{ given RED}$$
$$= \frac{2}{6} \times \frac{3}{20} = \frac{1}{20}$$

- The naive Bayes learner is trained by constructing a likelihood table for the appearance of these four words (W1, W2, W3,and W4), as shown in the following diagram for 100 emails

	Viagra (W_1)		Money (W_2)		Groceries (W_3)		Unsubscribe (W_4)		
Likelihood	**Yes**	**No**	**Yes**	**No**	**Yes**	**No**	**Yes**	**No**	**Total**
spam	4 / 20	16 / 20	10 / 20	10 / 20	0 / 20	20 / 20	12 / 20	8 / 20	20
ham	1 / 80	79 / 80	14 / 80	66 / 80	8 / 80	71 / 80	23 / 80	57 / 80	80
Total	5 / 100	95 / 100	24 / 100	76 / 100	8 / 100	91 / 100	35 / 100	65 / 100	100

- Using Bayes' theorem, we can define the problem as shown in the following formula,which captures the probability that a message is spam, given that **Viagra = Yes,Money = No, Groceries = No, and Unsubscribe = Yes:**

$$P(\mathrm{Spam} \mid W1\ \mathrm{I}\ \neg W2\ \mathrm{I}\ \neg W3\ \mathrm{I}\ W4) = \frac{P(W1\ \mathrm{I}\ \neg W2\ \mathrm{I}\ \neg W3\ \mathrm{I}\ W4 \mid \mathrm{spam})\, P(\mathrm{spam})}{P(W1\ \mathrm{I}\ \neg W2\ \mathrm{I}\ \neg W3\ \mathrm{I}\ W4)}$$

- Naive Bayes assumes class-conditional independence, which means that events are independent so long as they are conditioned on the same class value.

- P (Spam | W 1 I ¬ W 2 I ¬ W 3 I W 4) =
  P (W 1 | spam) P (¬ W 2 | spam) P (¬ W 3 | spam) P (W 4 | spam) P (spam)

  P (W 1) P (¬ W 2) P (¬ W 3) P (W 4)

- The result of this formula should be compared to the probability that the message is ham:
- $P (ham \mid W 1 \, I \, \neg W 2 \, I \, \neg W 3 \, I \, W 4) =$
- $P (W 1 \mid ham) P (\neg W 2 \mid ham) P (\neg W 3 \mid ham) P (W 4 \mid ham) P (ham)$
- ---
- $P (W 1) P (\neg W 2) P (\neg W 3) P (W 4)$

- The overall likelihood of **spam** is then:
- (4/20) * (10/20) * (20/20) * (12/20) * (20/100) = **0.012**
- While the likelihood of **ham** given this pattern of words is:
- (1/80) * (66/80) * (71/80) * (23/80) * (80/100) = **0.002**

- Because 0.012 / 0.002 = 6, we can say that this message is six times more likely to be spam than ham.
- The **probability of spam is equal to the likelihood that the message is spam divided by the likelihood that the message is either spam or ham:**
- 0.012 / (0.012 + 0.002) = **0.857**
- Similarly, the **probability of ham is equal to the likelihood that the message is ham divided by the likelihood that the message is either spam or ham:**
- 0.002 / (0.012 + 0.002) = **0.143**

Example

- Step 1: Convert the data set into a frequency table
- Step 2: Create Likelihood table by finding the probabilities like Overcast probability = 0.29 and probability of playing is 0.64.

Weather	Play
Sunny	No
Overcast	Yes
Rainy	Yes
Sunny	Yes
Sunny	Yes
Overcast	Yes
Rainy	No
Rainy	No
Sunny	Yes
Rainy	Yes
Sunny	No
Overcast	Yes
Overcast	Yes
Rainy	No

Frequency Table		
Weather	No	Yes
Overcast		4
Rainy	3	2
Sunny	2	3
Grand Total	5	9

Likelihood table				
Weather	No	Yes		
Overcast		4	=4/14	0.29
Rainy	3	2	=5/14	0.36
Sunny	2	3	=5/14	0.36
All	5	9		
	=5/14	=9/14		
	0.36	0.64		

Naive Bayes Example1

Table 1: Dataset

Outlook	Temperature	Humidity	Windy	Play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

- p(yes)=9/14
- p(no)= 5/14

Outlook	Temperature	Humidity	Wind	Play
Sunny	Cool	High	True	?

Based on yes probability:

$$P(\text{outlook} = \text{sunny} \,/\, \text{play} = \text{yes}) = \frac{2}{9}$$

$$P(\text{temperature} = \text{cool} \,/\, \text{play} = \text{yes}) = \frac{3}{9}$$

$$P(\text{Humidity} = \text{high} \,/\, \text{play} = \text{yes}) = \frac{3}{9}$$

$$P(\text{Wind} = \text{true} \,/\, \text{play} = \text{yes}) = \frac{3}{9}$$

$$P(\text{play} = \text{yes}) = \frac{9}{14}$$

Based on No Probability

p(sunny=yes/play=no)=3/5
p(temp=cool/play=no)=1/5
p(humidity=high/play=no)=4/5
p(wind=true/play=no)=3/5

- P(X/play= yes)P(play=yes)=
- X= Outlook, Temperature, Humidity, Wind

  P(X/play=yes)P(play=yes) = (2/9)*(3/9)*(3/9)*(3/9)*(9/14)

  P(X/play=yes)P(play=yes) = 0.0053

=.222*.333*.333*.333*.642=.00526

- P(X/play= no)P(play=no)=
- 3/5*1/5*4/5*3/5*5/14=.020
- p(no)>p(yes)

- **Play =no**

Example 2- In Syllabus

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

Frequency Table

		Stolen?	
		Yes	No
Color	Red	3	2
	Yellow	2	3

Likelihood Table

		Stolen?	
		P(Yes)	P(No)
Color	Red	3/5	2/5
	Yellow	2/5	3/5

Frequency Table

		Stolen?	
		Yes	No
Type	Sports	4	2
	SUV	1	3

Likelihood Table

		Stolen?	
		P(Yes)	P(No)
Type	Sports	4/5	2/5
	SUV	1/5	3/5

Frequency Table

		Stolen?	
		Yes	No
Origin	Domestic	2	3
	Imported	3	2

Likelihood Table

		Stolen?	
		P(Yes)	P(No)
Origin	Domestic	2/5	3/5
	Imported	3/5	2/5

- **p(color=red,type=suv,origin=domestic stolen=?)**
- p(stolen=yes)=5/10
- p(stolen=no)=5/10

- **Based on yes probability**
- p(color=red/stolen=yes)*p(type=suv/stolen=yes)*p(origin=domestic/stolen=yes)*p(stolen=yes) =
- 3/5*1/5*2/5*5/10=.024
- **Based on no probability**
- p(color=red/stolen=no)*p(type=suv/stolen=no)*p(origin=domestic/stolen=no)*p(stolen=no) =
- 2/5*3/5*3/5*5/10=.4*.6*.6*.5=.072
- p(stolen=no)>p(stolen=yes) , so p(color=red,type=suv,origin=domestic) is **not stolen**

Naive Bayes -Example Problem 3

headache	runny nose	fever	flu
N	Y	Y	N
Y	N	N	N
N	N	N	N
Y	Y	Y	Y
Y	Y	N	Y
N	N	Y	Y

headache	runny nose	fever	flu
Y	N	Y	?

chills	runny nose	headache	fever	flu?
Y	N	Mild	Y	N
Y	Y	No	N	Y
Y	N	Strong	Y	Y
N	Y	Mild	Y	Y
N	N	No	N	N
N	Y	Strong	Y	Y
N	Y	Strong	N	N
Y	Y	Mild	Y	Y

p(chills=y,runnynose =n,headache =mild,fever =y) ? flu =y / n???

- **Yes case probabilities**
- **P(flu=y)=5/8**
- p(chills=y/flu=y)=3/5
- p(runnynose=n/flu =y)=1/5
- p(headache=mild/flu=y)=2/5
- p(fever=y/flu=y)=4/5

- **No case probabilities**
- **P(flu=no)=3/8**
- p(chills=y/flu=no)=1/3
- p(runnynose=n/flu=n)=2/3
- p(headache=mild/flu=no)=1/3
- p(fever=y/flu=no)=1/3

- p(x/yes)=3/5*1/5*2/5*4/5*5/8=0.6*0.2*0.4*.8*.625=.024
- p(x/no)=1/3*2/3*1/3*1/3*3/8=.333*.666*.333*.333*.375=.0092
- p(x/yes)>p(x/no)
- So ,**p(chills=y,runnynose =n,headache =mild,fever =y) will be flu=yes**

Probabilistic Learning

- In machine learning, a probabilistic classifier is a classifier that is able to predict, given an observation of an input, a probability distribution over a set of classes, rather than only outputting the most likely class that the observation should belong to.

The Laplace estimator

- Let's look at one more example. Suppose we received another message, this time containing the terms: Viagra, Groceries, Money, and Unsubscribe. Using the naive Bayes algorithm as before, we can compute the likelihood of spam as:
- (4/20) * (10/20) * (0/20) * (12/20) * (20/100) = 0
- And the likelihood of ham is:
- (1/80) * (14/80) * (8/80) * (23/80) * (80/100) = 0.00005
- Therefore, the probability of spam is:
- 0 / (0 + 0.0099) = 0
- And the probability of ham is:
- 0.00005 / (0 + 0. 0.00005) = 1
- These **results suggest that the message is spam with 0 percent probability and ham with 100 percent probability.**

- A solution to this problem involves using something called the Laplace estimator,which is named after the French mathematician Pierre-Simon Laplace.
- The Laplace estimator essentially adds a small number to each of the counts in the frequency table, which ensures that each feature has a nonzero probability of occurring with each class. Typically, the Laplace estimator is set to 1 , which ensures that each class-feature combination is found in the data at least once.

- The overall likelihood of **spam** is then:
- (4/20) * (10/20) * (0/20) * (12/20) * (20/100) = 0
- While the likelihood of **ham** given this pattern of words is:
- (1/80) * (14/80) * (8/80) * (23/80) * (80/100) = 0.00005

- The likelihood of spam is therefore:
- (5/24) * (11/24) * (1/24) * (13/24) * (20/100) = 0.0004
- And the likelihood of ham is:
- (2/84) * (15/84) * (9/84) * (24/84) * (80/100) = 0.0001

Using numeric features with naive Bayes

- One easy and effective solution is to discretize numeric features, which simply means that the numbers are put into categories known as bins.
- For this reason, discretization is also sometimes called binning. This method is ideal when there are large amounts of training data, a common condition when working with naive Bayes.

Data Discretization

Data discretization refers to a method of converting a huge number of data values into smaller ones so that the evaluation and management of data become easy. In other words, data discretization is a method of converting attributes values of continuous data into a finite set of intervals with minimum data loss.

Example

Age

1,5,9,4,7,11,14,17,13,18, 19,31,33,36,42,44,46,70,74,78,77

After Discretization

Attribute	Age	Age	Age	Age
	1,5,4,9,7	11,14,17,13,18,19	31,33,36,42,44,46	70,74,77,78
After Discretization	Child	Young	Mature	Old

Another example is analytics, where we gather the static data of website visitors

Some Famous techniques of data discretization

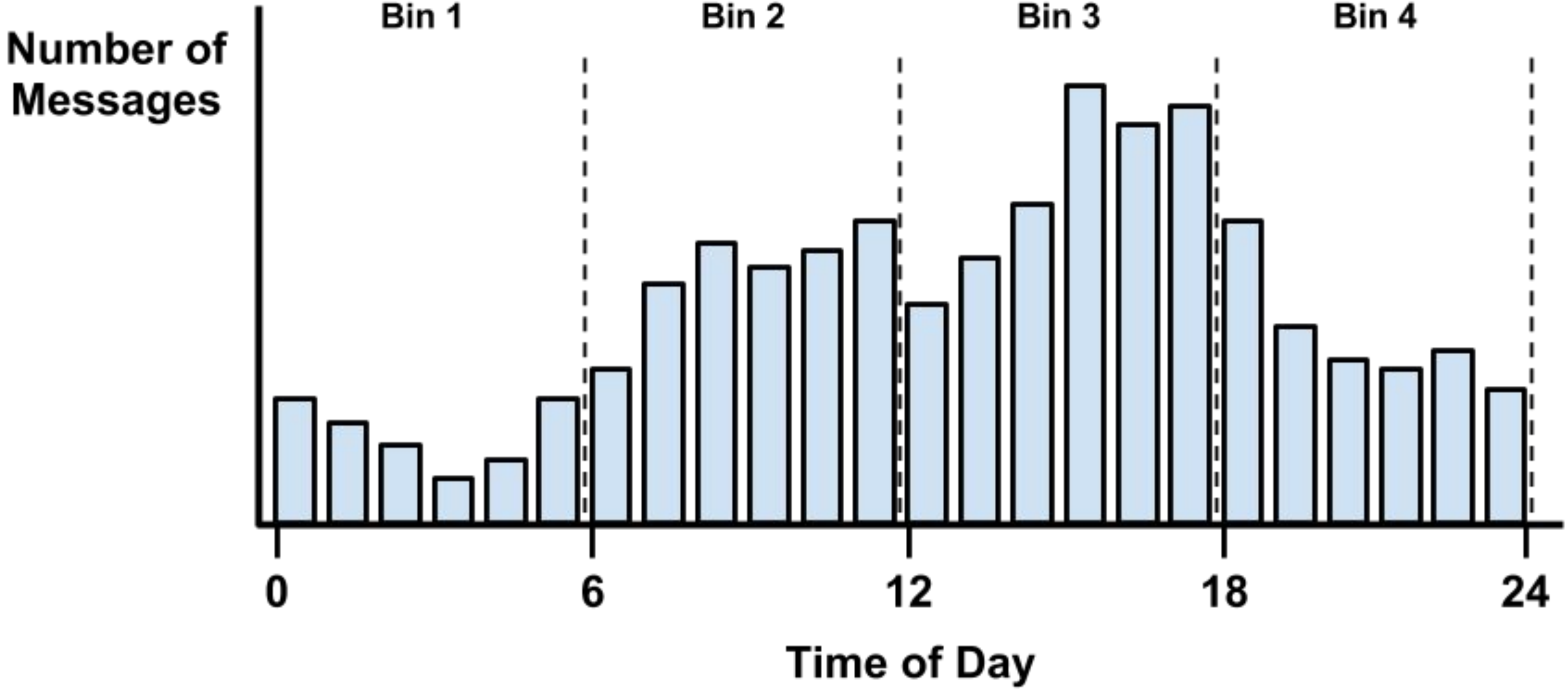
Histogram analysis

Histogram refers to a plot used to represent the underlying frequency distribution of a continuous data set. Histogram assists the data inspection for data distribution. For example, Outliers, skewness representation, normal distribution representation, etc.

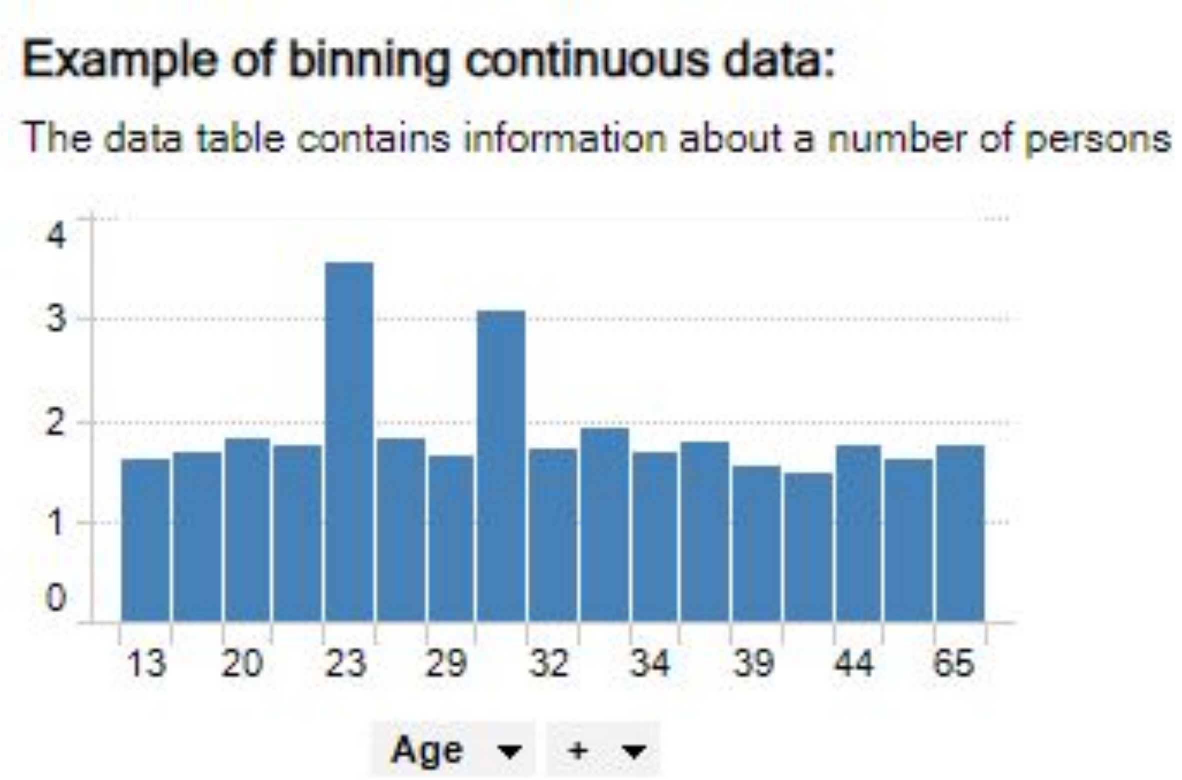
Binning

Binning refers to a data smoothing technique that helps to group a huge number of continuous values into smaller values. For data discretization and the development of idea hierarchy, this technique can also be used.

- For example, suppose that you added a feature to the spam dataset that recorded the time of night or day the email was sent, from 0 to 24 hours past midnight.
- Activity picks up during business hours, and tapers off in the evening. This seems to create four natural bins of activity, as partitioned by the **dashed lines indicating places where the numeric data are divided into levels of a new nominal feature, which could then be used with naive Bayes:**


Number of Messages
Bin 1
Bin 2
Bin 3
Bin 4
0
6
12
18
24
Time of Day

Before Binning



After Binning

Naive Bayes Example Problem 6

Table 8.1 Class-Labeled Training Tuples from the *AllElectronics* Customer Database

RID	*age*	*income*	*student*	*credit_rating*	*Class: buys_computer*
1	youth	high	no	fair	no
2	youth	high	no	excellent	no
3	middle_aged	high	no	fair	yes
4	senior	medium	no	fair	yes
5	senior	low	yes	fair	yes
6	senior	low	yes	excellent	no
7	middle_aged	low	yes	excellent	yes
8	youth	medium	no	fair	no
9	youth	low	yes	fair	yes
10	senior	medium	yes	fair	yes
11	youth	medium	yes	excellent	yes
12	middle_aged	medium	no	excellent	yes
13	middle_aged	high	yes	fair	yes
14	senior	medium	no	excellent	no

Classify X =(age =youth, income =medium, student =yes, credit rating =fair)

Naive Bayes -Example 7

Type of family structure	Age group	Income status	Will they buy a car?
Nuclear	Young	Low	Yes
Extended	Old	Low	No
Childless	Middle-aged	Low	No
Childless	Young	Medium	Yes
Single Parent	Middle-aged	Medium	Yes
Childless	Young	Low	No
Nuclear	Old	High	Yes
Nuclear	Middle-aged	Medium	Yes
Extended	Middle-aged	High	Yes
Single Parent	Old	Low	No

Classify (Single Parent, Young,Low)

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?